

A cross-sectional equity system for the NSE, and what it is worth net of costs

Machine-learning models rank 500 Indian stocks each day; the strategy buys the best-ranked tenth and sells short the worst. Over 9.3 years the ranking has genuine predictive skill. None of the six committed configurations is profitable enough to pay for the trading it requires.



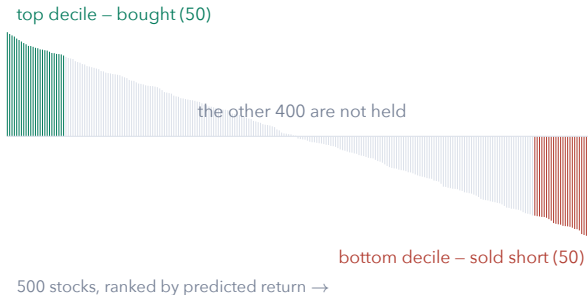
Growth of Rs 1 in the best of the six configurations (long only, 21-day target), before and after measured transaction costs. Log scale. 2016-01-21 to 2025-06-27.

Ranking, not forecasting

The system does not try to predict whether the market will rise or fall. Each trading day it ranks all 500 stocks against one another by predicted return over the next few days.

It then buys the top 10% and *sells short* the bottom 10% – short selling means borrowing a share, selling it, and buying it back later, which profits when the price falls.

Because the strategy is long and short at the same time, a general market move affects both sides and largely cancels. What is left depends on whether the *ordering* was right.



Schematic. One day's cross-section ordered by predicted return; the per-name scores are not retained on disk, so the shape is drawn rather than measured.

Ranking, not forecasting – *continued*

| This also means the strategy can make money in a falling market and lose money in a rising one. It is a bet on relative performance only.

Five stages, each a function of the one before

Tier 0	Tier 1	Tier 2	Tier 3	Tier 4
Data load prices, validate, forward-fill	Trees 17 features → three models → one score	Regime calm vs. stressed; de-risk gate	Execution weights, costs, realized return	Gate is the result credible?

Five stages, each a function of the one before – *continued*

TWO RULES HOLD EVERYWHERE

Missing prices are **carried forward, never backward**.

Filling a gap with a later price would let the strategy act on information that did not exist yet.

A signal formed on day T earns the return from day T to T+1. It is never credited with the move that produced it.

WHY THE LAST STAGE EXISTS

A backtest can always be made to look good by trying configurations until one works. Stage 4 measures how many were tried and raises the bar accordingly, so that a result which is merely the luckiest of many attempts does not pass.

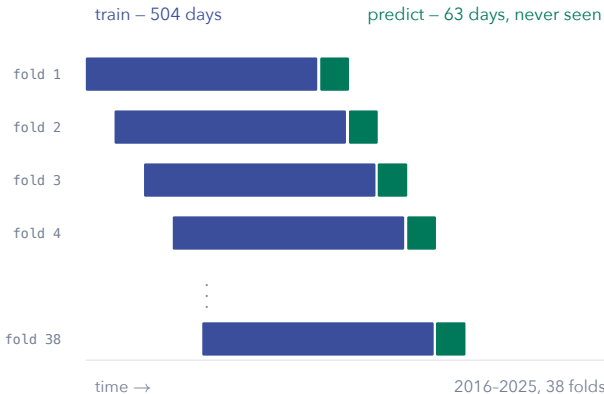
The models are never tested on what they were taught

A model that is fitted and scored on the same period will always look skilful. This one is refitted repeatedly through history.

Two years of data train three models – LightGBM, XGBoost and a random forest. They then predict the following quarter, which they have never seen. The window advances and everything is refitted.

Over the sample this happens **38 times**. Every scored day was predicted by a model fitted only on days before it.

| The three models' predictions are averaged.
Averaging different learners is more stable than trusting any single one.

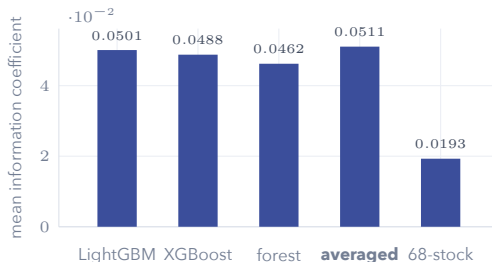


Walk-forward estimation. Each row is one refit: the training window on the left, the out-of-sample quarter it then predicts on the right.

The ranking does predict

Skill is measured by the **information coefficient**: on each day, the correlation between the predicted ordering and what actually happened. Zero means the ranking is noise. In equities, 0.02–0.05 is a genuinely useful signal – the bar is low because so little of a single stock's daily move is predictable at all. The three-model average reaches **0.0511**, and ranks in the right direction on **69%** of days across 2,333 days.

This is the finding the rest of the deck has to be read against. **The signal is real.** Everything that follows is about what it costs to act on it.



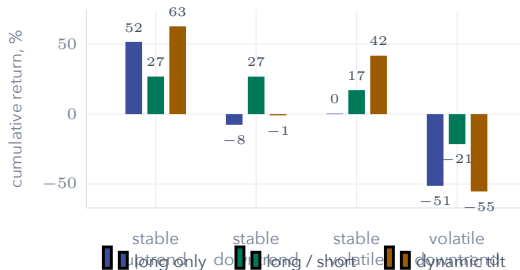
Mean daily cross-sectional rank IC by model, 500-name universe, 5-day target, 2,333 scored days. The rightmost bar is the earlier 68-stock universe for scale.

Phases 1 and 2 — the apparatus

Phase 1 ran the whole five-stage stack on *simulated* prices for ten stocks, three of which were given a hidden upward drift. It confirmed that the machinery works end to end.

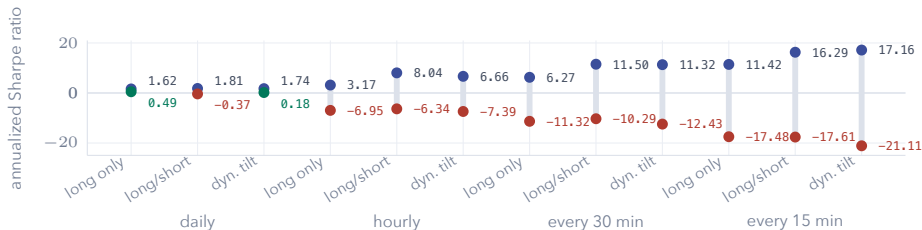
It says nothing about markets. The simulator decided which stocks would win, so finding them demonstrates only that the search works – which is exactly what it was built to check.

Phase 2 replaced the simulation with real NSE data, replaced the simple momentum ranking with the tree models, and removed a look-ahead flaw in the regime detector, which had been fitted on the whole history at once.



Phase 1, cumulative return by simulated market regime and trading style.
Synthetic data: a verification of pipeline mechanics, not an empirical result.

Phase 3 — the first collapse



Each line runs from a configuration's Sharpe before costs to its Sharpe after costs. Gross values from the Phase 3 record; net values recomputed from the 12-cell ledger. 68 stocks, 2018-2025.

Phase 3 charged the actual Indian statutory cost of trading – securities transaction tax, exchange and regulator fees, stamp duty, GST and brokerage – about **14.7 basis points each time a position is opened or closed** (a basis point is one hundredth of one percent).

Phase 3 — the first collapse – *continued*

Faster trading means more of those charges. Every intraday configuration went to **-100%**: the strategy trades so often that the fees exceed the edge with certainty, not by bad luck. The only survivor was the daily long-only book, at a Sharpe ratio of 0.57 – a measure of return per unit of risk, where above 1 is good and below 0 loses money – which still fell short of the credibility gate.

Widening the universe from 68 stocks to 500

Standard portfolio theory says that skill of a given quality applied to more independent bets produces a better risk-adjusted return, scaling roughly with the square root of the number of bets. Going from 68 names to 500 predicts an improvement of about **2.7x**, against the **1.5x** needed to pass.

Testing that required rebuilding the dataset from the exchange's public daily archive as a **point-in-time** index membership: a company that belonged to the index in 2017 and later collapsed is still present in 2017.

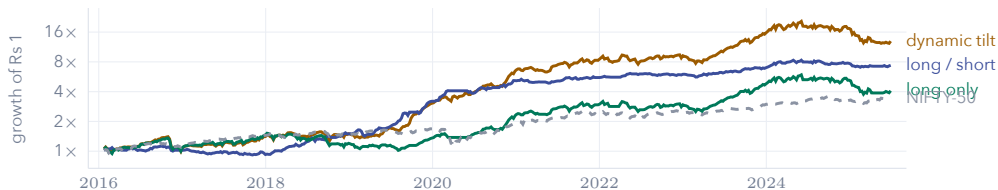


Membership churn per year, computed from the point-in-time universe mask. Names entering above the line, names leaving below.

Widening the universe from 68 stocks to 500 – *continued*

Without this, a backtest quietly restricts itself to firms that survived – which is information nobody had at the time. Over the sample, **1,035 different companies** passed through a universe that only ever holds 500 at once.

Phase 4b — two of three configurations passed



Growth of Rs 1, computed from the Phase 4b ledger; index from the exchange archive. Log scale, so equal vertical distances are equal percentage gains. Costs charged here are statutory only.

On the widened universe the strategy performed as the breadth argument predicted, and better. The long/short book returned **639.2%** over 9.3 years at a Sharpe of **1.78**, against **252.3%** for the NIFTY-50 index. Two of the three configurations cleared the credibility gate.

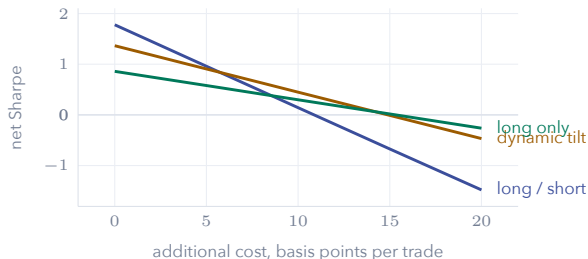
Phase 4b — two of three configurations passed – *continued*

This result was recorded as **preliminary rather than accepted**. Three quantities the backtest relied on had not been measured; they are the subject of the next slide, and of the phase that followed.

CONFIGURATION	TOTAL RETURN	SHARPE	WORST DRAWDOWN	GATE
long only	308.4%	0.86	−36.3%	0.916
long short	639.2%	1.78	−19.9%	1.000
dynamic tilt	1194.7%	1.37	−41.6%	0.998
NIFTY-50 index	252.3%	0.91	−38.4%	—

Three quantities the backtest had not measured

1. **The cost of actually transacting was set to zero.** Only the statutory taxes and fees were charged. The bid-ask spread – the gap between the price to buy and the price to sell – and the price movement caused by one's own order were both absent.
2. **Nobody had checked the short positions could be borrowed.** Shorting in India requires the stock to have a futures contract or an available lender. The worst-ranked names are precisely where that is scarce.
3. **The trial count was understated.** The credibility gate was told three configurations had been tried. The true figure was far higher.



Sharpe against additional trading cost, in basis points per transaction. The passing configurations stop passing at about 6.8 and 7.8 basis points – less than $2\times$ headroom on a quantity that had not been measured. Horizontal axis: basis points.

Three quantities the backtest had not measured – *continued*

| The first of these decides the outcome. Each additional basis point of trading cost removes roughly 2.1% a year from the long/short book, because it turns over most of its positions every day.

Phase 4c — measuring all three, and proving nothing else changed

Track A

Measure the cost

half-spread estimated from
daily prices; market impact
charged per name

Track B

Verify the shorts

point-in-time borrowable set
from traded futures
contracts

Track D

Correct the statistics

honest trial count $N = 30$;
Lo correction for serial
correlation

Phase 4c — measuring all three, and proving nothing else changed – *continued*

THE CONTROL THAT MAKES THE COMPARISON VALID

Every addition was written so that it is **switched off by default**. With the switches off, the earlier runs were re-executed and had to reproduce their previous output before any new figure was accepted.

They reproduced to the last decimal: a maximum difference of 1.1×10^{-16} across all three Phase 4b configurations, which is the rounding error of the arithmetic itself, and exactly 0.0 on the Phase 3 ledger.

WHY THAT MATTERS

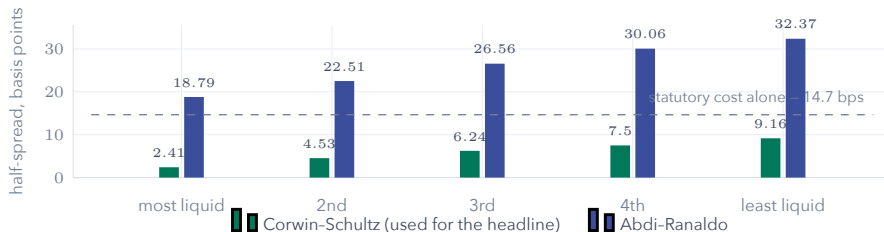
Rewriting a backtest and getting a different answer leaves two possible explanations: the new measurement, or a mistake introduced while rewriting.

Because the old configuration still returns the old numbers exactly, every difference reported from here on is attributable to the cost and eligibility models, and not to the rewrite.

| A further rule: a strategy restricted to borrowable shorts is treated as a *different* strategy with its own entry in the trial count, not as a corrected version of the old one. Running both and reporting the better is the error the gate exists to catch.

RE-RUN WITH THE NEW MEASUREMENTS OFF	COLUMNS	LARGEST DIFFERENCE FROM THE PUBLISHED RESULT
Phase 3, daily, 5-day target	3	0.0 – bit for bit
Phase 4b, 500 names, 5-day target	3	1.1×10^{-16} – floating-point rounding

Track A — measuring the spread that had been assumed away



Estimated half-spread by liquidity, both estimators, 500-name universe. From the Phase 4c measurement record (docs/phase4c_results.md §1).

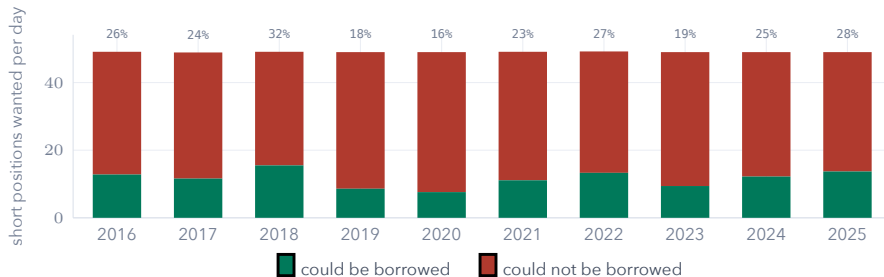
The daily exchange archive records no bid and ask prices, so the spread has to be *estimated* from the open, high, low and close. Two published estimators were used and kept as a range, because they are known to err in opposite directions: Corwin-Schultz reads low at **5.97 bps** and Abdi-Rinaldo reads high at **26.07 bps**.

Track A — measuring the spread that had been assumed away – *continued*

The strategy grid was run on the **lower** estimate deliberately. If the result fails under the generous assumption, the conclusion cannot be attributed to a pessimistic choice of estimator.

Both estimators rise monotonically as stocks become less liquid, which was the condition set in advance for trusting them. Once charged, the real cost of a transaction is about **26 bps** against the **14.7 bps** of statutory charges – and the long/short Sharpe falls from **1.78 to –0.23**.

Track B — most of the short book could not have been borrowed



Intended short positions per year, split by whether the name was borrowable. From the Phase 4c feasibility record (docs/phase4c_results.md §6); the in-universe eligible count independently recomputes to 36% here.

Whether a stock can be sold short was determined from the exchange's derivatives archive: a name is treated as borrowable on a given day only if futures contracts in it actually traded that day. Because it is derived from real trading, it carries no hindsight.

About a third of the universe qualifies. But the strategy does not want a random third – it wants the **50 worst-ranked names**, and those are systematically the ones no lender offers.

Track B — most of the short book could not have been borrowed – *continued*

Averaged over the sample the strategy intended to short **49 names** a day and could have borrowed **11**. Coverage never exceeds 32% in any year. Roughly **78% of the short book was never available**, and because futures eligibility is more generous than actual stock lending, even that is an upper bound.

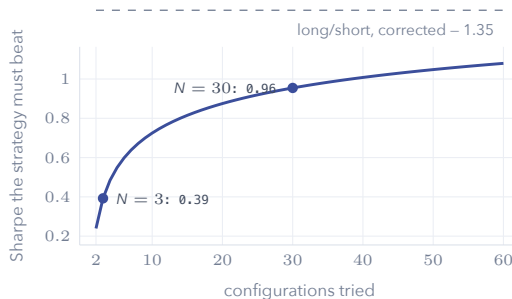
Restricting the strategy to what it could have borrowed cuts the long/short book's Sharpe *before costs* from **4.20 to 1.88**. More than half the raw signal lived in positions that could not have been taken.

Track D — counting the attempts honestly

If thirty strategies are tested, the best of them will look impressive even when none has any skill, simply because it is the maximum of thirty draws. The gate corrects for this by raising the Sharpe a strategy must beat as the number of attempts grows.

Phase 4b had used **$N = 3$** . Phase 4c replaced this with a line-by-line tally of every configuration the project had ever run: **$N = 30$** . The bar rises from **0.39** to **0.98**.

A second correction was applied for the fact that overlapping multi-day forecasts make consecutive returns related to one another, which flatters the Sharpe ratio.

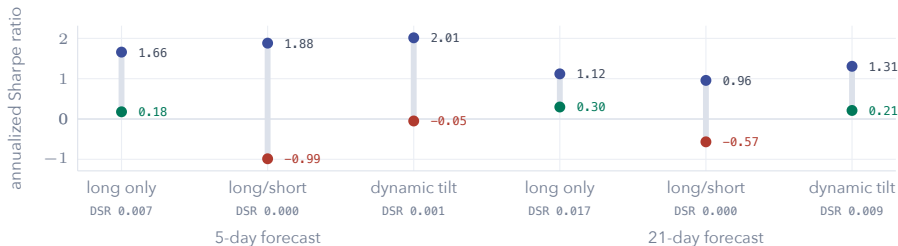


The deflation benchmark as a function of the number of configurations searched, computed with the repository's own estimator at the measured cross-trial dispersion. Horizontal axis: configurations tried.

Track D — counting the attempts honestly – *continued*

| Applied on their own, the statistical corrections do **not** overturn Phase 4b – the long/short book still passes at 0.986.
The change of verdict comes from the measured costs.

The committed grid — none of the six passes



Sharpe before and after measured costs for each committed configuration, with its credibility score. Computed from the Phase 4c ledger and gate output.

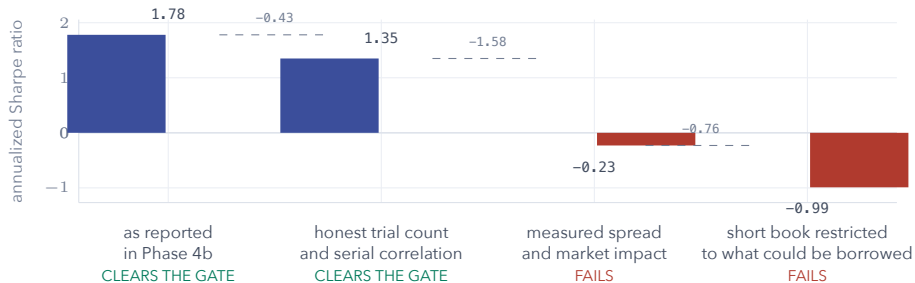
Six configurations were fixed in writing *before* the run: three trading styles, each with a five-day and a twenty-one-day forecast horizon. Fixing them in advance is what stops the search from continuing until something works.

All six fail. The best reaches a credibility score of **0.017** against a threshold of **0.95** – under 2% of the confidence required, not a near miss. For every one of the six, the minimum track record needed to reach credibility is **undefined**: none clears the bar at any length of history, so collecting more data would not change the answer.

The committed grid — none of the six passes – *continued*

The slower forecast horizon did what it was intended to do – turnover fell from **0.45 to 0.24** of the book per day and the net Sharpe rose from **0.18 to 0.30**. It was the one remaining lever with room in it, and it moved the result by 0.12 against a gate that needed far more.

What each correction was worth



The long/short configuration, Phase 4b to Phase 4c, one correction at a time. Vertical axis: annualized Sharpe ratio.

Findings



Significance of the daily ranking by year, from the Phase 4c diagnostics. A value near 2 is the conventional threshold.

THE SIGNAL IS REAL, AND IT IS NOT A SMALL-STOCK ILLUSION

Skill is flat across the liquidity spectrum – an information coefficient of 0.041 in the most traded fifth of the universe and 0.048 in the least. The leading suspicion, that the edge was an artefact of thinly traded names, is ruled out. The strategy fails because a genuine 0.04 does not cover 26 basis points a side while replacing a quarter to three quarters of the book every day.

THE SIGNAL IS FADING

Its statistical significance falls from about 10 in 2016–2019 to **1.21** in 2025, at which point it is indistinguishable from zero.

Findings

THE BREADTH ARGUMENT IS SETTLED, NEGATIVELY

Widening the universe from 68 names to 500 produced the predicted improvement before costs. It also raised turnover from 0.32 to 0.45 of the book per day and extended the strategy into stocks with wider spreads. Breadth bought signal and cost together.

WHAT THIS DOES NOT RULE OUT

A portfolio construction that treats cost as part of the objective rather than a charge applied afterwards; and neutralizing market and sector exposure, deliberately excluded here because it adds a degree of freedom that requires its own committed prior and its own trial count.

| Each correction was applied to every configuration alike and regardless of whether it improved the result. That is the basis on which these figures are reported.